

## Pixel Response Times: The Long And Short Of It

**R**esponse Time can be defined as the amount of time it takes a pixel to go from its active state to its passive state and again back to its active state. Unlike a CRT, where the pixel response time is virtually negligible, LCDs have been plagued with high response times—you end up getting ghosting on your screen because the transition time (switching from on to off) is so high.

The liquid crystals in an LCD display actually move, or re-orient themselves accordingly, when electricity is passed through them. The time taken by these viscous crystals to re-orient is a few milliseconds, and the movement could be minute or large depending on the colour the pixel is supposed to display. The Off position for a pixel is white, while the On position is black. One complete pixel operation would therefore be a shift from white to black to white. In such a case, remember that the distance the crystal has to move is maximum, since black represents the maximum amount of orientation away from white (other colours will fall in between). So technically, the

crystal would take the longest possible time for this operation, and therefore the response time would be the slowest, right?

Wrong! You have to consider that the angle or degree of turning is proportionate to the intensity of the electrical field applied. Therefore to orient the crystal just a little bit, a very weak electrical field is needed, and the slower the crystal moves. The larger the degree of orientation required, the greater the electricity applied, and the faster the crystal will turn, and vice-versa—a rather paradoxical situation. In many multimedia applications, games for example, the response time for a pixel transition from, say, black to dark grey may be more significant than the transition from black to white.

Transitions from grey to grey are the fastest, and so most manufacturers quote this figure. For a monitor having a “marketed” response time of 4 ms gtg (grey to grey), the actual response time may be closer to 8 ms or even 12 ms. The ISO standard for pixel latency is defined as the time taken for a full black to white transition.

times, but suffered from poor colour reproduction. S-IPS panels had amazing colour reproduction and acceptable response times, but suffered from poor contrast ratios. MVA panels walked the middle path to try and address all flaws in previous technologies, and offer actual contrast ratios of 600:1, which translates to blacks actually looking black, and a full spectrum of grey. So do we finally have visual perfection?

Unfortunately, no! MVA panels have been proven to have the worst pixel response times. Also, if a close comparison is done, MVA panels

do not offer the same brilliant colour reproduction that S-IPS panels do, although to be pretty honest, MVA panels leave TN based ones chewing the dust on this parameter.

MVA panels are usually natively 8-bit. However, the drive for economy has seen cheaper 6-bit MVA panels that use dithering, similar to TN panels.

Once again, MVA panels are expensive, much more so than TNs, but cheaper than IPS panels, so they’re mostly restricted to around the 19-inch-and-above category.



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